MODEL CARD · V1.0 · NLP CA3

# GreenLens

Greenwashing Claim Auditor — an explainable NLP classifier for environmental marketing claims

**Task** Text classification | **Labels** greenwashing / not_greenwashing | **Data** EmeraldData · 620 claims · CC-BY-4.0

| 620 | 225 / 395 | 434 / 186 | 0.625 |
|---|---|---|---|
| claims audited | greenwashing / not | train / held-out test | best F1 (ML, held-out) |

## 1 Model Description

### Rule-based lexicon

Scores vague terms, unsupported superlatives, broad-scope wording and checkable evidence markers (numbers, dates, certifications). Transparent by construction.

### ML · TF-IDF + Logistic Regression

1–2 gram features. Trained on **434** claims, tested on a held-out split of **186** it never saw.

**Intended use:** screening claims for human review. Not for legal findings or automated enforcement.

## 2 Overall Performance

| 0.556 | 0.625 |
|---|---|
| Rule-based F1 · all 620 | ML F1 · held-out 186 |

| Model | Acc | Prec | Rec | F1 |
|---|---|---|---|---|
| Rule-based lexicon | 0.694 | 0.586 | 0.529 | 0.556 |
| ML (TF-IDF + LogReg) | 0.742 | 0.667 | 0.588 | 0.625 |

Evaluation sets differ, so the comparison is indicative, not like-for-like.

## 3 Bias Check — Performance Across Subgroups

**A · Claim length** (short vs longer statements)




| Subgroup | N | Acc | Prec | Rec |
|---|---|---|---|---|
| Short | 333 | 0.724 | 0.741 | 0.581 |
| Long | 287 | 0.659 | 0.379 | 0.429 |

**B · Quoted slogans vs. plain statements**




| Subgroup | N | Acc | Prec | Rec |
|---|---|---|---|---|
| Quoted slogan | 105 | 0.581 | 0.700 | 0.681 |
| Plain statement | 515 | 0.717 | 0.526 | 0.458 |

**Finding:** weaker on quoted slogans (58.1% vs 71.7%) and long multi-clause claims (65.9% vs 72.4%). Long claims often bury a vague aspiration inside an otherwise factual sentence, which the lexicon under-weights.

## 4 Explainability — Top Features

Terms most associated with the **greenwashing** class (LogReg coefficients):

`million` `power` `2021` `electric` `program` `aims` `2040` `plans`

Not only vague words like "eco-friendly": future-tense commitments ("aims", "plans", a target year like 2040) get flagged more than completed, dated actions. Common pattern: **a future target or small pilot presented as an achieved, company-wide result.**

## 5 Worked Example

**"Amazon powers 100% of its company activities with renewable energy."**

FLAGGED confidence 78%

**Evidence:** no date or completion qualifier. **Pattern:** absolute claim ("100%") with no "by [year]".

**Contrast:** "on track to power 100%… by 2025" is correctly **left unflagged**.

## 6 Limitations

**1 · Semi-synthetic data.** Claims and labels were LLM-generated and annotated, anchored to real ESG reports/news, not scraped from live ads. Revalidate before trusting absolute numbers on real ad copy.

**2 · Register bias.** Underperforms on quoted, slogan-style language, the exact register real ads use.

**3 · Not a legal or factual determination.** "Flagged" means the claim resembles greenwashing patterns in this dataset, not proof of lying. Route to human review, not automated enforcement.



Data: EmeraldData (geoka/EmeraldData) · CC-BY-4.0